Kwikby_MP Improvement Plan

A plan for improving the code in github.com/fadim1965/Kwikby_MP. This is a plan only: no code or files in the repository were changed. It is based on a read-only clone and a survey of the files.

What I found

- **Already in the repo:** the stock consolidation (`StockSync` is used in 8 files), the demo seeder, the EF `InitialSchema` migration, and about 395 tests.
- **Written but not wired in:**
  - `ProductionGuards` exists and is never called from `Program.cs`.
  - `ProductImageProcessor` has no callers.
  - The stored procedures and hero engine in `Database/` are not referenced by any app code.
- **Security gaps:**
  - `appsettings.json` commits `Seed:AdminPassword = Admin#12345` and `DemoShops = true`.
  - `AddDataProtection()` has no key persistence, and the app sets no cookie `Secure` policy and no forwarded headers.
- **Clutter and structure:**
  - There are two solution files.
  - Sample data sits at the repo root: `images/`, `csv/`, `ready/`, `stock-usages.txt`.
  - `docs/` holds about 8 MB of old LibanPost PDFs.
  - There is no CI, `.editorconfig` or `.gitattributes`.
  - 323 files still use the `WebDS_Test` namespace, and 56 files mention LibanPost.
  - Six C# files exceed 900 lines, and `ThemeCatalog.cs` has 2,176.

Plan, in order

Phase	Work	Risk
0. Hygiene	Delete `WebDS_Test.sln`. Move sample data, `ProductImageRules.md` and the gap analysis out of the root, into `samples/` or `docs/`. Drop the generated PDFs, the empty `package.json` and the Bootstrap `.map` files from git. Add `.gitattributes`, `.editorconfig` and a CI workflow that builds and tests.	Low
1. Security and config	Remove the committed admin password, and default demo shops to off. Require a private admin password outside Development, and make the demo customer's password random. Call `ProductionGuards` from `Program.cs`. Persist Data Protection keys, add forwarded headers and a cookie `Secure` policy, and fix the per-policy `Retry-After`. Force a password change on first sign-in.	Medium
2. Data and schema	Make EF migrations the only schema source. Apply them at startup and baseline existing databases. Retire the `*Schema.cs` DDL helpers and `00_Schema.sql`. Fold the earlier schema fixes into a migration. Add a database check that stock is never negative. Decide what to do with the unused stored procedures.	Medium to High
3. Photos and import	Connect the photo processor, and store photos under `shops/{ShopId}/products/`. Add `accept="image/png"` to the forms. Let the import create missing categories, and add a row limit. Replace the admin form's silent "first shop" default with a shop picker.	Medium

Phase	Work	Risk
4. Code quality	Rename the namespace and branding in one isolated commit. Move `ThemeCatalog` data out of C#. Split `OrderService`, `PostSaleService`, `ReportingService` and `AccountController` by responsibility. Extract a database initialiser from `Program.cs`. Turn on analyzers. Add SQL Server integration tests for checkout, stock concurrency, webhook idempotency and shop suspension.	Medium
5. Product gaps	Review the checkout, payment and ledger code, which has not been read in this version of the repo. Add a real payment provider, translate the new English labels, check right-to-left layout, add health checks, and integrate the hero engine.	Needs input

Each phase would be a separate branch with tests, so any change can be reverted on its own.

Decisions needed

1. **Stored procedures and hero engine:** integrate them, or move them to `docs/proposals` until needed?
2. **Upload location:** `App_Data/uploads` (survives redeploys) or `wwwroot/uploads`? The public URL is the same either way.
3. **SQL Server version:** stay on 2014, or upgrade? Several scripts need 2016 SP1.
4. **Namespace rename:** do it now, before more code lands, or later?
5. **Git history:** leave it alone? The committed password is the public demo one, so rewriting history is only worth it if the repo goes public.

Caveats

- The project has not been built and no tests were run.
- The counts come from searching the files, so treat them as approximate.
- Checkout, payments and returns have not been reviewed in this version of the repo, so Phase 5 is a placeholder.